Accelerated Math II/III Name_____

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HW #3 - Polynomials & Linear Factors

Date_____ Period____

Find the zeros of the function. Then graph the function using your graphing calculator to check your solutions.

1) $f(x) = x(x+3)(x+1)$

2) $f(x) = x(x+4)(x-1)$

3) $f(x) = x(2x+5)(x-1)$

4) $f(x) = x(2x-1)^2$

5) $f(x) = x(2x+3)^2$

6) $f(x) = x(2x+1)(2x-5)$

Write each function in factored form. (Hint: Use what you learned in the previous unit about factoring)

7) $f(x) = 3x^3 - 7x^2 - 6x$

8) $f(x) = 2x^3 - 3x^2 + x$

Sketch the graph of each function. Approximate each real zero to the nearest tenth. Approximate the relative minima and relative maxima to the nearest tenth. (You will need to use your graphing calculator.)

9) $f(x) = x^3 - 2x^2 - x + 2$

10) $f(x) = 2x^3 + 3x^2 - 1$

Sketch the graph of each function. Approximate each real zero to the nearest tenth. Approximate the relative minima and relative maxima to the nearest tenth. (You will need to use your graphing calculator.)

11) $f(x) = x^3 + 11x^2 + 35x + 28$

12) $f(x) = x^3 + x^2 - x + 1$

13) $f(x) = x^3 - x^2 - 3$

14) $f(x) = -x^4 - x^3 + 4x^2 - 2$

Write a polynomial function of least degree with integer coefficients that has the given zeros.

15) 0, 5, −5

16) 2, $-\frac{3}{2}$, $-\frac{2}{5}$

17) −3 mult. 2, 1, 3

18) −2 mult. 2, 4

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Answers to HW #3 - Polynomials & Linear Factors

1) $\{0, -3, -1\}$

2) $\{0, -4, 1\}$

3) $\left\{0, -\frac{5}{2}, 1\right\}$

4) $\left\{0, \frac{1}{2} \text{ mult. } 2\right\}$

5) $\left\{0, -\frac{3}{2} \text{ mult. } 2\right\}$

6) $\left\{0, -\frac{1}{2}, \frac{5}{2}\right\}$

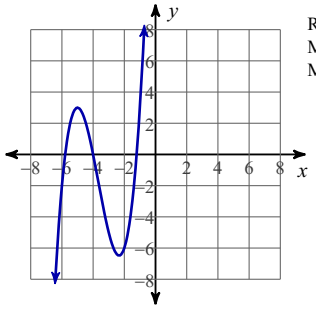
7) $f(x) = x(3x + 2)(x - 3)$

8) $f(x) = x(2x - 1)(x - 1)$

9) $\{2, 1, -1\}$

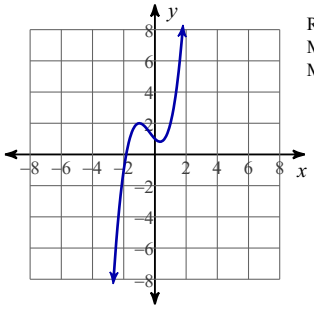
10) $\left\{\frac{1}{2}, -1 \text{ mult. } 2\right\}$

11)



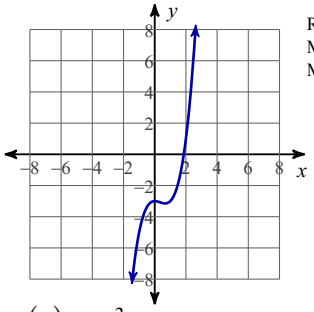
Real Zeros: −5.8, −4, −1.2
Minima: (−2.3, −6.5)
Maxima: (−5, 3)

12)



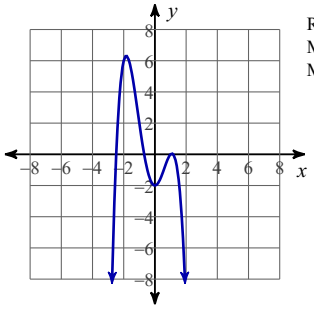
Real Zeros: −1.8
Minima: (0.3, 0.8)
Maxima: (−1, 2)

13)



Real Zeros: 1.9
Minima: (0.7, −3.1)
Maxima: (0, −3)

14)



Real Zeros: −2.5, −0.7, 1, 1.2
Minima: (0, −2)
Maxima: (−1.8, 6.3)
(1.1, 0)

15) $f(x) = x^3 - 25x$

16) $f(x) = 10x^3 - x^2 - 32x - 12$

17) $f(x) = x^4 + 2x^3 - 12x^2 - 18x + 27$

18) $f(x) = x^3 - 12x - 16$

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